

AGA Clinical Practice Update on Endoscopic Resection for Early Colorectal Cancer: Commentary



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DESCRIPTION: T1 colorectal cancer (CRC), defined as tumor invasion confined to the submucosa, has traditionally been managed by surgery. Advances in endoscopic resection techniques, coupled with an improved understanding of tumor biology and risk of lymph node metastasis and recurrence, have expanded our treatment armamentarium for these lesions. The purpose of this American Gastroenterological Association (AGA) Clinical Practice Update Expert Commentary is to review the definition of T1 CRC, discuss the role of endoscopic resection for these lesions, review factors associated with risk of recurrence and metastasis, and discuss post-resection management, including surveillance or adjuvant surgery. The target audience for this commentary includes primary care, gastroenterology, pathology, and surgical health care professionals. **METHODS:** This expert commentary was commissioned and approved by the AGA Institute Clinical Practice Updates Committee (CPUC) and the AGA Governing Board to provide timely guidance on a topic of high clinical importance to the AGA membership and underwent internal peer review by the CPUC and external peer review through standard procedures of *Gastroenterology*.

Keywords: Colorectal Cancer; Adenocarcinoma; Endoscopic Mucosal Resection; Endoscopic Submucosal Dissection.

Definition of Early Colorectal Carcinoma

Colorectal cancer (CRC) is the third most common cancer in the United States, with an annual incidence rate of 42.9 per 100,000 people.¹ As per the National Comprehensive Cancer Network guidelines,² CRC is defined clinically by invasion of neoplastic cells through the muscularis mucosae into the submucosa. Without submucosal invasion (SMI), the term *cancer* should be avoided. Terms such as *intramucosal adenocarcinoma* and *in situ adenocarcinoma*, which refer to lesions in which the dysplastic changes are limited to the lamina propria alone, should be avoided, as the inclusion of the word *adenocarcinoma* can result in misinterpretation and mismanagement.

Early CRC (T1 cancer), sometimes called a “malignant polyp,” is defined as a lesion with SMI but without invasion into the muscularis propria. Detailed optical lesion assessment is the first critical step in the evaluation of these lesions. Although a comprehensive discussion of optical imaging classifications is outside the scope of this commentary, clinicians should recognize the key morphologic features (eg, depressed/excavated lesion, irregular pit/

vascular pattern, and lateral spreading granular mixed or nongranular lesions) commonly associated with a higher risk for occult cancer, as they provide important information to direct management decisions (Figure 1).³ Nonetheless, reliance on optical diagnosis alone has its drawbacks, as real-time optical diagnosis of T1 CRC is highly endoscopist-dependent, and its performance outside of trials and specialized academic centers has been conflicting.⁴ Indeed, the diagnostic performance of established optical classification systems has shown limited accuracy in the detection of cancer among both Asian and non-Asian endoscopists alike, with relatively low sensitivity in the 40%–50% range.⁵ Lesion misclassification based on optical diagnosis can potentially result in unnecessary surgery for otherwise endoscopically resectable lesions. Hence, when available, we suggest referral to a high-volume center with expertise in endoscopic resection and pathology when the diagnosis is unclear and when there is a high level of suspicion for CRC. Careful endoscopic tattooing (so that the tattoos do not spread underneath the lesion) may be performed to enable subsequent endoscopic and surgical localization of the lesions, unless the lesions are located in the cecum or lower rectum. For lesions with suspicious features of deep SMI, such as Narrow-Band Imaging International Colorectal Endoscopic Classification (NICE) type 3 or excavated lesions, biopsy and consideration for surgery is a reasonable strategy.⁶ However, the risk of potentially underdiagnosing cancer based on biopsies alone, due to inadequate tissue sampling and interpretation, should not be underestimated.⁷ Hence, endoscopic local excision, when technically feasible, can also be considered as a viable strategy for histologic diagnosis with the potential of curative intent, particularly when patients who undergo additional surgery after noncurative endoscopic resection have long-term outcomes similar to those who undergo primary surgery for T1 CRC.⁸

Abbreviations used in this paper: CRC, colorectal cancer; EFTR, endoscopic full-thickness resection; ESD, endoscopic submucosal dissection; ESGE, European Society of Gastrointestinal Endoscopy; JGES, Japan Gastroenterological Endoscopy Society; JSCCR, Japanese Society for Cancer of the Colon and Rectum; LNM, lymph node metastasis; LVI, lymphovascular invasion; SMI, submucosal invasion; USMSTF, US Multi-Society Task Force.

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0016-5085/\$36.00

<https://doi.org/10.1053/j.gastro.2025.09.001>

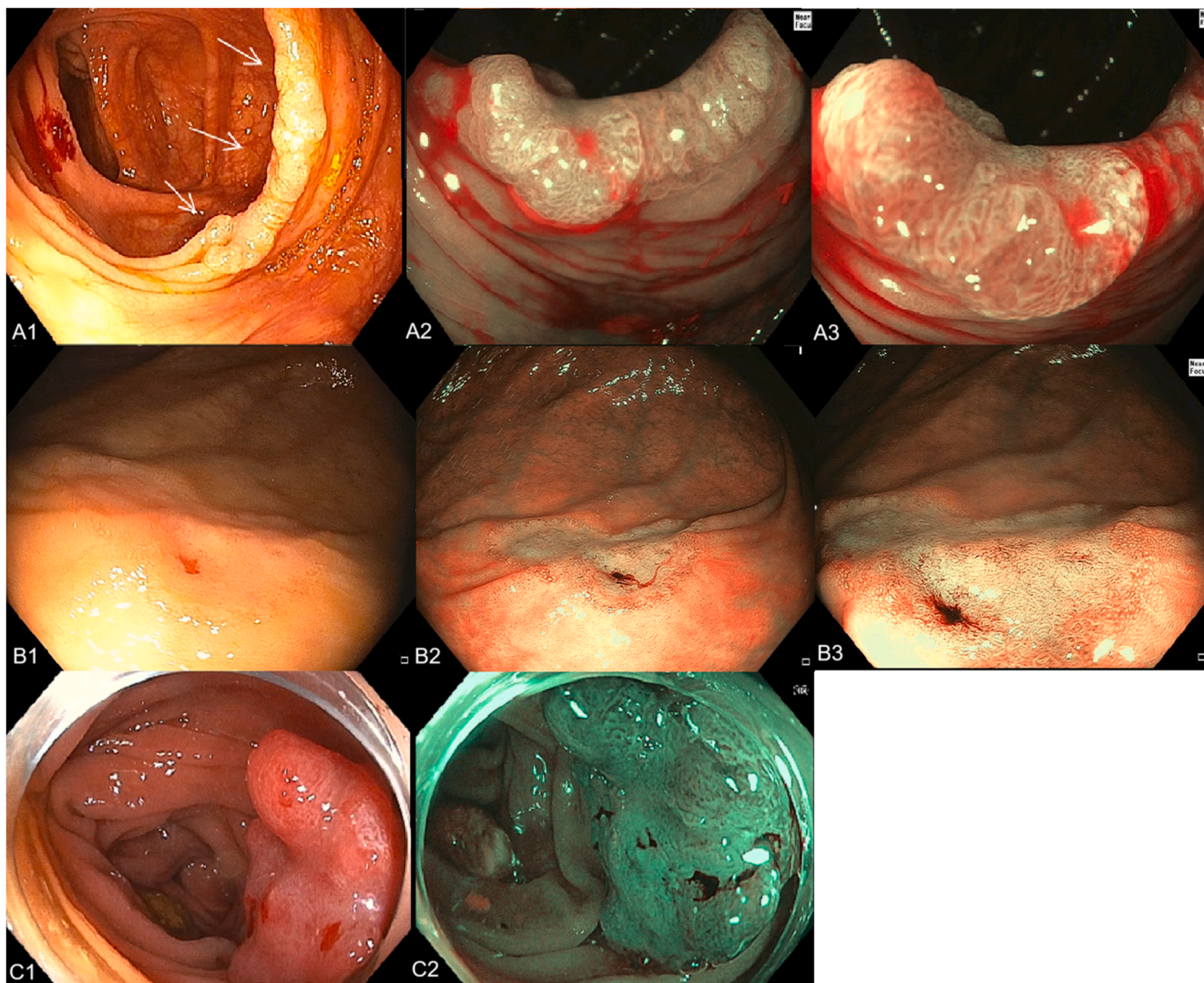


Figure 1. Endoscopic features suggestive of SMI: (A) Lateral spreading nongranular tumor with pseudodepression (*arrows*) in the ascending colon. (A1) White light images. (A2, A3) Narrow band imaging (NBI). (B) Lateral spreading nongranular tumor with pseudodepression in the cecum. (B1) White light images. (B2) NBI. (B3) NBI with near-focus mode. (C) Sessile lesion in the sigmoid colon. (C1) White light images. (C2) NBI shows NICE type III surface pattern.

Endoscopic Resection of T1 Colorectal Cancer

As outlined by the US Multi-Society Task Force (USMSTF), European Society of Gastrointestinal Endoscopy (ESGE), and Japan Gastroenterological Endoscopy Society (JGES), suspected T1 CRC should ideally be approached with en-bloc resection.^{9–11} En-bloc removal is indispensable, as piecemeal resection hinders accurate histologic assessment. Histologic evaluation is not only less reliable in the setting of a fragmented specimen, but piecemeal resection is also an independent risk factor for luminal recurrence. The National Comprehensive Cancer Network practice guidelines specify that T1 lesions removed in piecemeal fashion by endoscopy will inevitably require

adjuvant surgery due to the potential risk of understaging the lesion because of compromised pathologic interpretation and the risk of local recurrence.²

Given the higher en-bloc resection rate with endoscopic submucosal dissection (ESD) compared with endoscopic mucosal resection, irrespective of lesion size, ESD is often the preferred strategy for suspected superficially invasive T1 CRCs. Nonetheless, ESD is limited by its technical complexity, steep learning curve, and inability to confidently resect lesions with deep SMI. Endoscopic full-thickness resection (EFTR) may be a reasonable safe and effective alternative, particularly for select T1 CRCs <2 cm in size in which deep SMI is suspected.^{12,13} According to the USMSTF, ESGE, and JGES, endoscopic resection is considered curative when the following criteria are met: (1)

resection is complete with histopathologic margins free of tumor, (2) superficial SMI ($<1000\ \mu\text{m}$), (3) absence of lymphovascular invasion (LVI), (4) well to moderately differentiated histology (G1/G2), and (5) low or absent tumor budding.^{9–11}

It should be noted that the criteria defining high-risk pT1 CRC have a variable impact on the incidence of lymph node metastasis (LNM) and recurrence. In the following section, we further elaborate on these histopathologic features and their impact on endoscopic resection and additional decision making, including the need for adjuvant surgery and multidisciplinary evaluation.

Factors Associated With Lymph Node Metastasis

Depth of Submucosal Invasion

The depth of SMI is commonly measured in micrometers using the method recommended by the Japanese Society for Cancer of the Colon and Rectum (JSCCR).¹⁴ For nonpedunculated lesions, depth of SMI should be measured from the bottom of the muscularis mucosae or from the surface of the lesion when the muscularis mucosae is not identifiable to the invasive front of the tumor. For pedunculated lesions, the depth of SMI should be estimated based on the Haggitt classification.¹⁵ This system categorizes lesions into the following 5 classes: level 0, neoplastic cells confined to the mucosa; level 1, invasion into the submucosa, but limited to the head of the polyp; level 2, invasion extends into the neck of the polyp; level 3, invasion into any part of the stalk; and level 4, invasion into the submucosa of underlying bowel wall below the stalk, but above the muscularis propria.¹⁵

Deep SMI ($\geq 1000\ \mu\text{m}$) has traditionally been identified as a risk for LNM and an important criterion for surgery in T1 CRC.^{9–11} However, emerging data suggest that metastatic risk based on depth of SMI alone, in the absence of other high-risk features, may be lower than previously anticipated. Indeed, it is estimated that $>95\%$ of patients with deep SMI as the only criterion for “high-risk” T1 CRC do not have LNM. In a recent meta-analysis, Zwager et al¹⁶ found that SMI as a solitary risk factor (8 studies; 1146 patients) was associated with an absolute risk of LNM of 2.6% and pooled incidence rate of 2.83 (95% CI, 1.66–4.78). Notably, in this large meta-analysis, deep SMI was not an independent predictor of LNM and 37.9% of cancers with deep SMI did not feature other high-risk factors. These findings further support the potential role of organ preserving treatment strategies for a substantial group of T1 CRCs. The relatively low risk of LNM in select patients raises the question of who should be considered for adjuvant surgery. In a population-based cohort of 5170 patients with T1 CRC who underwent surgery, 30-day mortality was 1.7%, and 8.3% had severe adverse events requiring re-intervention.¹⁷ Therefore, with a growing number of patients with advanced age and comorbidities, alternative strategies should be explored when accounting for operative risk.

Histologic Type

All CRC specimens, particularly after endoscopic resection, should be carefully reviewed by dedicated gastroenterology expert pathologists and discussed in a multidisciplinary board. Compared with well to moderately differentiated adenocarcinoma, poorly differentiated (G3), undifferentiated (G4), signet ring cell, and mucinous carcinomas are associated with a worse prognosis due to higher rates of LNM and recurrence.^{18,19} In a meta-analysis of 67 studies including 21,238 patients, adverse histologic type was identified as an independent risk factor for LNM (odds ratio, 2.14).¹⁸ The JSCCR guidelines categorize poorly differentiated adenocarcinoma, signet ring cell carcinoma, or mucinous carcinoma as advanced histology and recommend surgery as additional treatment.¹⁴ Similarly, the USMSTF and ESGE consider “poor” or “undifferentiated” histology as a high-risk criterion.^{9,10} Nonetheless, accurate histologic classification can be challenging for multiple reasons. For one, there is no consensus on how to report histologic type when different levels of differentiation are present in the same specimen; the World Health Organization suggested taking into account the lowest level of differentiation, whereas the Japanese guidelines suggest using the predominant differentiation.²⁰ Furthermore, studies have shown that pathologic assessment of malignant polyps can vary significantly among observers.²¹ In addition, it has been increasingly recognized that in certain cases, a residual submucosal lifting agent on the resected specimen can appear as amorphous substance mimicking gastrointestinal mucin.²²

Resection Margins

The impact of margin positivity on risk of recurrence and metastasis is an area of ongoing investigation. R0 resection refers to complete resection of a lesion in which the surgical margins are microscopically free of tumor. For R0 resection, both horizontal and vertical margins should be uninvolved. The site of margin positivity (lateral/horizontal vs deep/vertical) has different implications. Lateral/horizontal margin positivity usually denotes risk of local recurrence, whereas positive vertical/deep margins is usually associated with an increased risk of deeper tumor invasion and LNM. However, there is no clear consensus on what constitutes “free resection margins.” Traditionally, a resection margin $<1\ \text{mm}$ has been considered a risk for residual cancer, with small observational studies indicating a higher rate of cancer at the time of surgery and recurrence in this cohort of patients.²³ However, a more recent multicenter study of T1 CRC suggested that the risk of local recurrence was not significantly higher among those with resection margins between 0.1 and 1 mm (0.8%; 95% CI, 0.1%–4.4%) vs those with $>1\ \text{mm}$ in the absence of tumor budding.²⁴ Both the USMSTF and JSCCR guidelines define positive margin as one in which carcinoma is identified at the submucosal margin of the resected specimen. The ESGE specifies that R0 resection should preferably have a vertical margin $\geq 1\ \text{mm}$, with those $<1\ \text{mm}$ free of tumor needing stricter follow-up or surgery if other poor prognostic

histologic features are present, such as tumor budding. Additional research is needed to better define what constitutes “margin positivity” and its significance on risk of recurrence and metastasis.

Lymphovascular Invasion

LVI refers to the presence of neoplastic cells in a space bordered by a defined endothelium (lymphatics or blood vessels) and is considered the strongest risk factor for LNM. In a systematic review of the predictive value of pathologic risk factors for LNM, LVI was identified as the strongest independent predictor (relative risk, 5.2; 95% CI, 4.0–6.8).¹⁸ Ideally, the reporting of LVI should distinguish between small and large vessel involvement, as the former is associated with a stronger association with LNM.²⁰ Immunostaining of lymphovascular endothelium has been found to improve inter-observer agreement²⁵; albeit its role beyond H&E staining for LVI has not been fully elucidated. Although immunostaining may enhance detection of lymphovascular involvement, it remains to be determined whether higher rates of LVI positivity translate into oncologic benefit or conversely result in overtreatment. Currently, the USMSTF, JSCCR, and ESGE unanimously recognize LVI as a high-risk criterion that should prompt surgery when present.^{9–11} Although not as well described in the endoscopic literature, perineural invasion is also considered a risk factor for LNM in patients with CRC.²⁶

Tumor Budding

Tumor budding is defined as a single tumor cell or a cell cluster consisting of ≤ 4 tumor cells at the invasive margin of the polyp.²⁷ Tumor budding is stratified into the following 3 grades: 0–4 buds, low budding (Bd 1); 5–9 buds, intermediate budding (Bd 2); and 10 or more buds, high budding (Bd 3). In pT1 CRC, the presence of Bd 2 and Bd 3 is associated with an increased risk of LNM. Similar to LVI, the USMSTF, JSCCR, and ESGE recognized tumor budding as a high-risk criterion, with the meta-analysis by Zwager et al¹⁶ showing an odds ratio of 2.83 for risk of LNM.^{9–11,14} However, more research is needed on how to standardize staining methods to optimize histologic visualization and reporting.

Surveillance After Endoscopic Resection of High-Risk and Low-Risk T1 Colorectal Cancer

An algorithm following endoscopic resection of T1 CRCs is shown in Figure 2.²⁸ The overall risk of recurrence is low after endoscopic curative resection of T1 CRC without high-risk criteria. In a meta-analysis of 5167 endoscopically treated patients with T1 CRC,²⁹ the pooled incidence of recurrence (local or distant CRC) was 0.7% for low-risk (curative resection) and 7.0% for high-risk (noncurative resection) T1 CRC. Most CRC recurrences (60%) occur within 36 months and the vast majority (96%) of recurrences occurred within 72 months of follow-up. Furthermore, most of the recurrences for low-risk T1

CRCs are local and the risk of LNM appears to be relatively low. These data suggest that surveillance colonoscopy alone without monitoring for lymph nodes or distant metastasis is a reasonable strategy after curative resection of low-risk T1 CRC; although long-term prospective data are needed.

There is no clear evidence to inform optimal surveillance intervals after curative endoscopic resection of T1 CRCs. van Lanschot et al³⁰ reported that metachronous CRCs and recurrence were found in 1.7% of patients at surveillance colonoscopy 1 year after curative surgical CRC resection. Both the JSCCR and ESGE recommend surveillance colonoscopy within 1 year after endoscopic resection.^{10,14} However, it should be highlighted that outcomes after local surgical excision (and, by extension, endoscopic en-bloc resection) of early-stage rectal cancers, may be worse than for colon cancer and, as a result, sooner and more frequent surveillance after endoscopic resection of a low-risk T1 rectal cancer may be reasonable. Indeed, the National Comprehensive Cancer Network recommends surveillance colonoscopy at 1 year after endoscopic curative resection of colon cancer, but suggests more frequent surveillance for rectal cancer.² Additional information on surveillance after pathologic curative ESD of early gastrointestinal neoplasia can be found in a recent separate American Gastroenterological Association Clinical Practice Update.³¹

For patients with high-risk T1 CRC, surgery is the standard of care. However, mounting evidence suggests that the different prognostic factors may not weigh equally on the risk of LNM, and the decision to proceed with surgery should be individualized, balancing the potential risk of LNM and recurrence with the morbidity of surgery. For instance, high-risk rectal T1 CRCs treated by endoscopic resection have a significantly higher local recurrence rate and incidence of LNM compared with the colon lesions.³² Yet, rectal surgery has higher morbidity compared with other locations in the colon; thereby advocating the need for less invasive options.

In patients with local excision of a T1 CRC with R1/Rx (incomplete/indeterminate) resection margins without other histologic risk factors, the risk of intramural residual cancer and risk of LNM is 5%–11% and 8%–9%, respectively.¹⁹ Both the ESGE and JSCCR guidelines recommend endoscopic surveillance between 3 and 6 months for those with only lateral margin positivity without other high-risk features.^{6,8,20} Conversely, the National Comprehensive Cancer Network recommends adjuvant surgery in the presence of margin positivity, irrespective of location (lateral, vertical, or both).

Emerging data on local excision of the previous resection scar—with EFTR, transanal minimally invasive surgery for rectal lesions, or combined endoscopic-laparoscopic surgery for colonic lesions—has been proposed as an alternative strategy to oncologic surgery.¹² In a large retrospective study reporting long-term outcomes (median follow-up of 64 months) of patients with T1 CRC with R1/Rx resection margins as a sole risk factor, CRC recurrence was seen in 2.2% of patients ($n = 7$) who underwent adjuvant surgery and 9.0% of patients ($n = 8$) who underwent EFTR. Disease-free survival was lower in the full-thickness resection group (96.8% vs 89.9%; $P = .019$), but

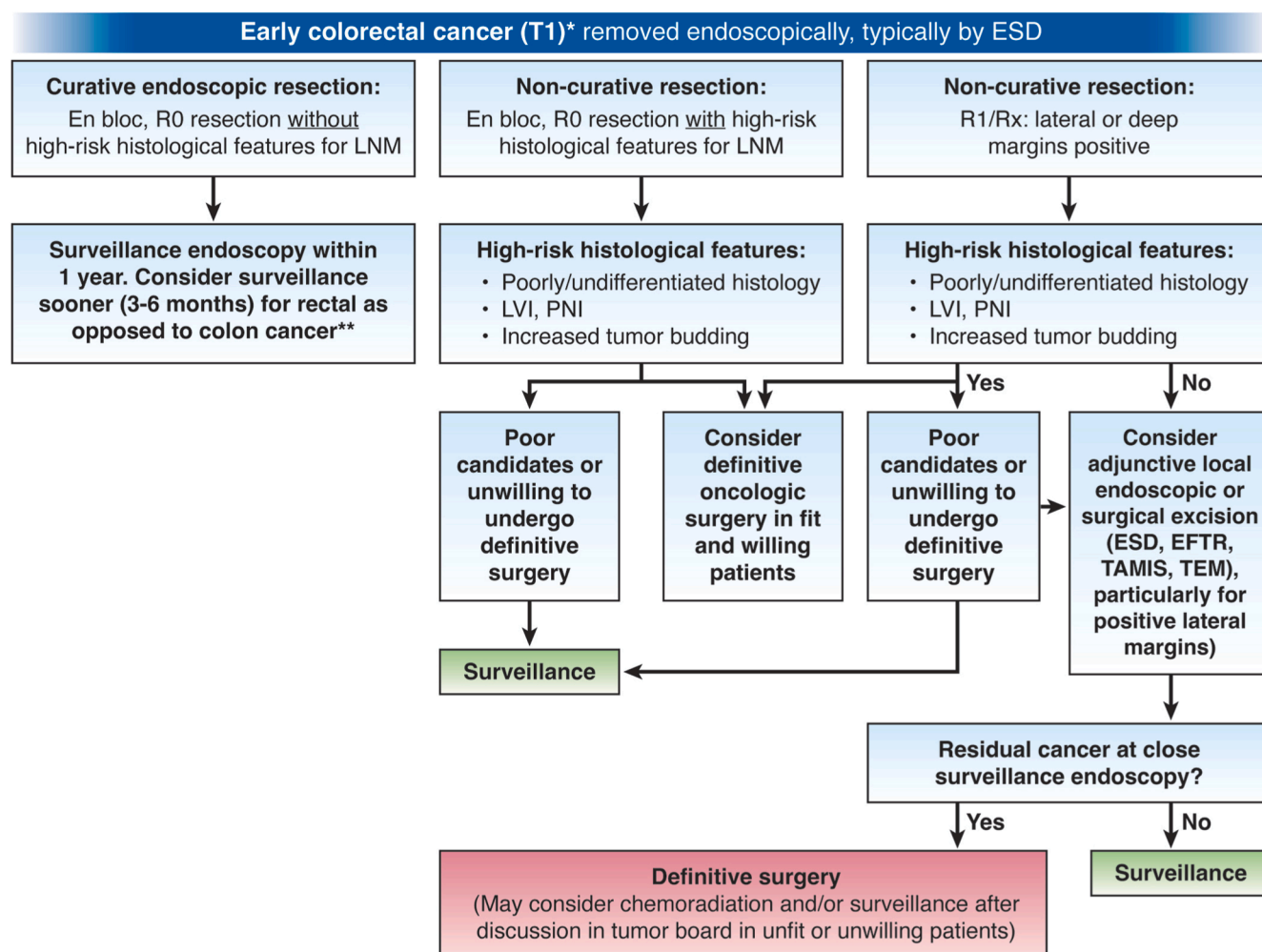


Figure 2. Suggested surveillance for malignant colorectal lesions removed by endoscopic resection. *Does not apply to hereditary cancer syndromes or inflammatory bowel disease. **Refer to Zwager et al.³² PNI, perineural invasion; TAMIS, transanal minimally invasive surgery; TEM, transanal endoscopic microsurgery.

5 of the 8 recurrences were adequately managed with adjuvant surgery, resulting in a similar 5-year metastasis-free survival and overall survival between the 2 groups.¹² These data suggest that full-thickness resection of the scar with a surveillance strategy could be an alternative to adjuvant surgery in the absence of other histologic risk factors; however, long-term data are needed, with prospective studies currently underway.³²

In summary, future research is warranted to explore individualized surveillance protocols based on risk stratification among patients with endoscopically treated T1 CRC. Notably, most of the data on endoscopic resection of T1 CRC originates from Europe and Asia. Additional studies are urgently needed to consider potential regional and racial differences when applying these findings to patient populations in North America.

Conclusions

The optimal approach to the management of T1 CRC remains complex and is evolving. The decision to proceed

with surgery should be individualized, balancing the perceived risk of LNM and recurrence with the morbidity of surgery. There are several limitations to the current accepted prognostic factors, including that all of these rely on histologic assessment and do not weigh equally on the risk of LNMs. Despite the use of standardized criteria, >80% of the cases deemed high risk do not have evidence of LNM at the time of surgery.³³ Considering the ongoing struggles to accurately categorize lesions and the need for histopathology for risk stratification, en-bloc endoscopic resection, specifically by using ESD, could be considered as an initial approach for select cases, which serves as both a diagnostic and a potentially curative approach. This suggested pathway seems reasonable, given the lower morbidity and mortality in comparison with surgery, particularly when long-term outcomes after secondary surgery for noncurative endoscopic resections are similar to those who undergo primary surgery.³⁴ Nonetheless, ESD remains a technically complex procedure and may not be widely available in the community. Hence, additional techniques, including EFTR, may be a potential alternative

in selected T1 CRCs. Future efforts are needed to validate the efficacy and safety of endoscopic resection for T1 CRCs, refine the current selection criteria, and identify new stratification parameters to develop predictive models to help tailor patient care.

Video

To view an accompanying video interview with the authors, please visit the online version of the article at <https://doi.org/10.1053/j.gastro.2025.09.001>.

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Received April 3, 2025. Accepted September 3, 2025.

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Author Contributions

All authors contributed in the review of the literature, drafting of the manuscript, writing and critical appraisal of the final manuscript.

Conflicts of interest

The authors disclose the following: Saowanee Ngamruengphong is a consultant for Boston Scientific, Fujifilm, Neptune Medical, and Olympus. Mohamed O. Othman is a consultant for Olympus, Boston Scientific, Creo Medical, Lumendi, and ConMed; and he receives research support from Boston Scientific, AbbVie, Nestle, ConMed, and Lucid Diagnostic. Andrew Y. Wang owns publicly traded stock in GE HealthCare Technologies and Pfizer. Dennis Yang is a consultant for Boston Scientific, Olympus, Fujifilm, Medtronic, Microtech, 3D-Matrix, Neptune Medical, ERBE and Aspero Medical; and he receives research support from Boston Scientific.